

Structural defects classified as “Minor” still generate disproportionately high repair costs, suggesting that severity labels alone may not fully reflect operational or financial impact.

## Executive Summary

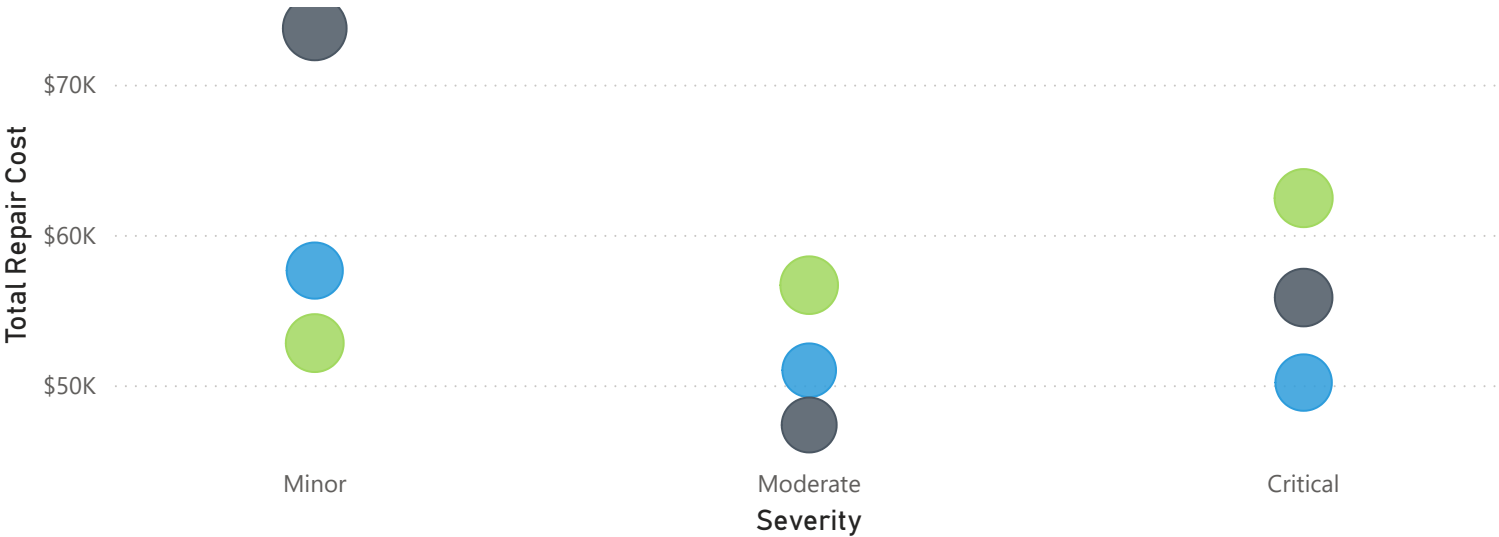
This analysis explores how defect severity, defect type, and inspection method influence repair cost exposure. The findings suggest that financial impact does not always align cleanly with severity classification alone.

The dataset used in this case study is synthetic and therefore more evenly distributed than many live operational datasets. The purpose of the analysis is to demonstrate how defect classification, financial impact, and operational exposure can be explored through structured reporting and visual analysis.

|                     |                                       |                                        |                             |                             |                                 |
|---------------------|---------------------------------------|----------------------------------------|-----------------------------|-----------------------------|---------------------------------|
| 1K<br>Total Defects | \$176.92K<br>Highest Defect Type Cost | Structural<br>Highest-Cost Defect Type | 33.3%<br>% Critical Defects | \$508K<br>Total Repair Cost | \$507.63<br>Average Repair Cost |
|---------------------|---------------------------------------|----------------------------------------|-----------------------------|-----------------------------|---------------------------------|

## Total Repair Cost and Total Defects by Defect Type and Severity

Defect Type    ● Cosmetic    ● Functional    ● Structural

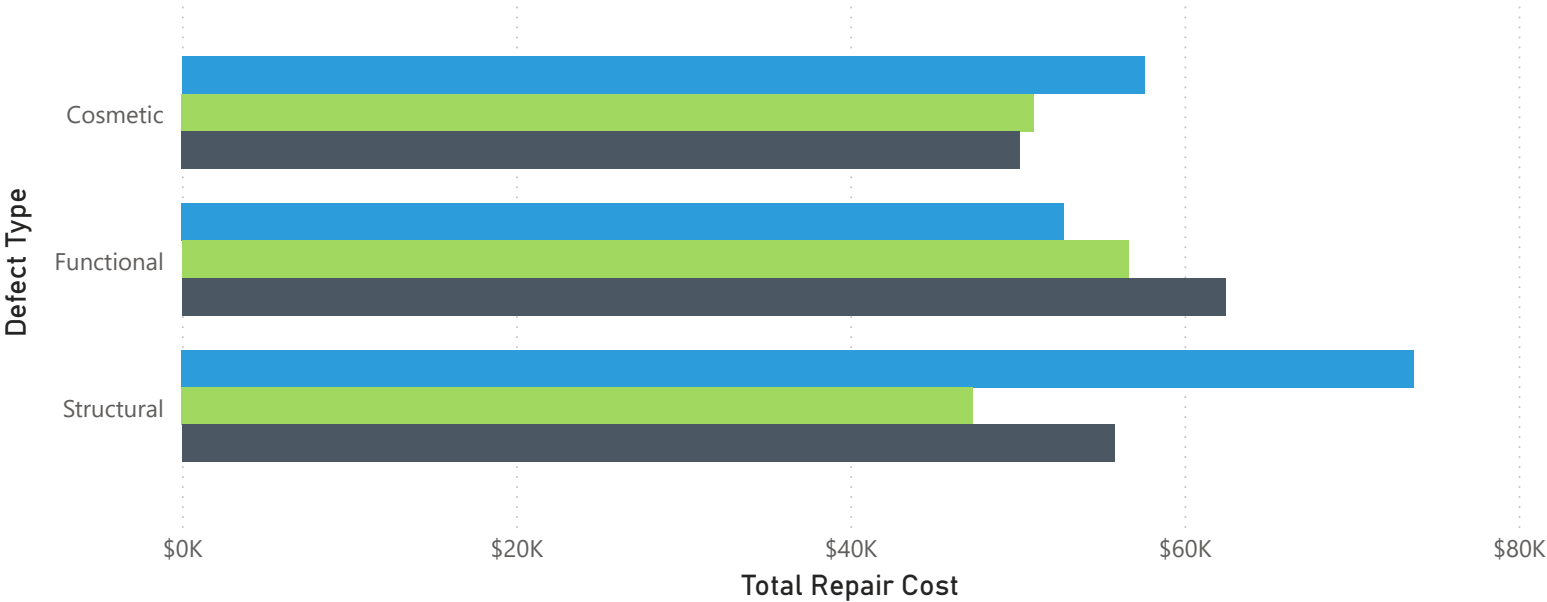


# Cost Concentration

Which defect categories create disproportionate repair exposure?

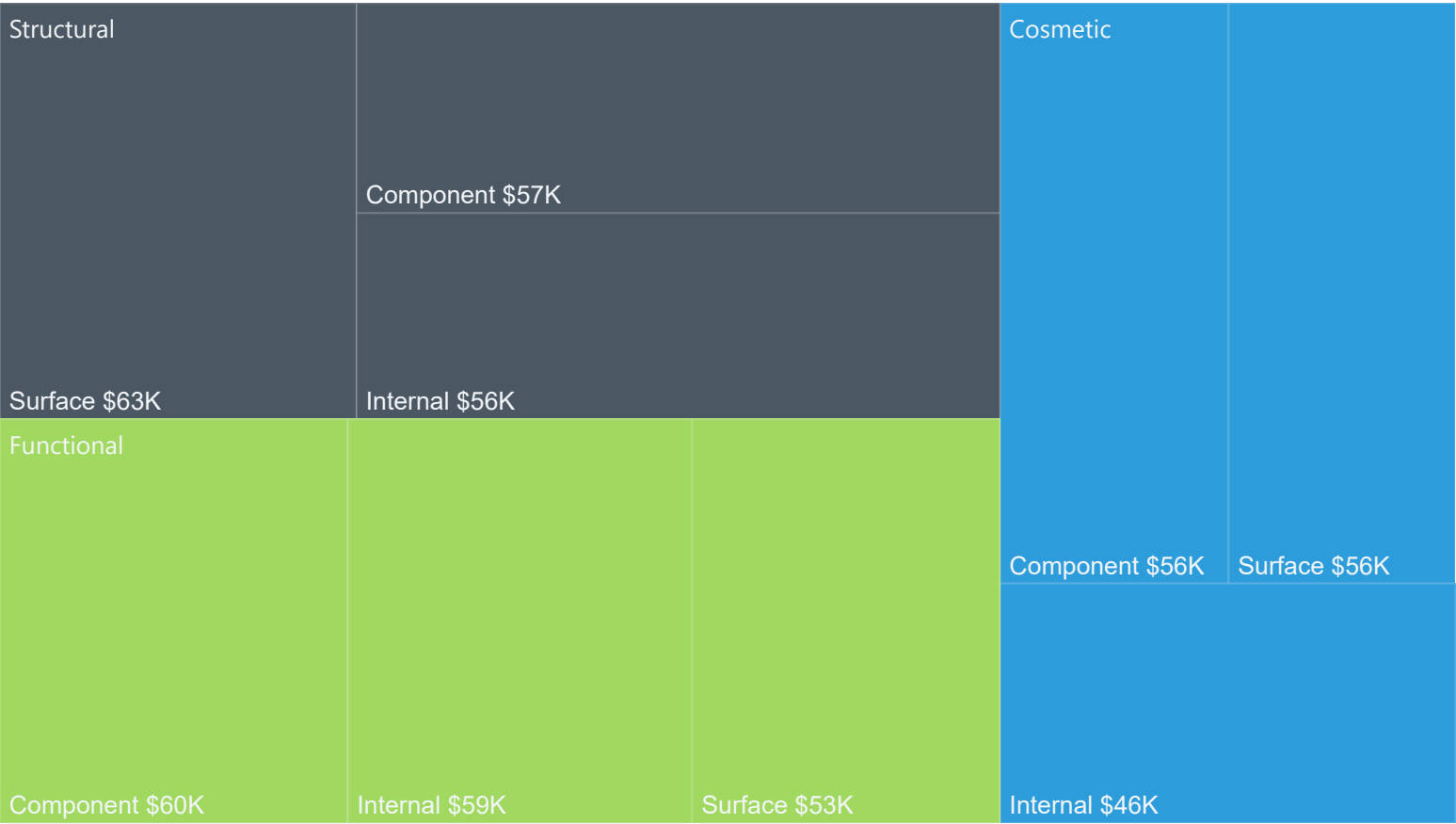
## Total Repair Cost by Defect Type and Severity

Severity Minor Moderate Critical



## Total Repair Cost by Defect Type and Defect Location

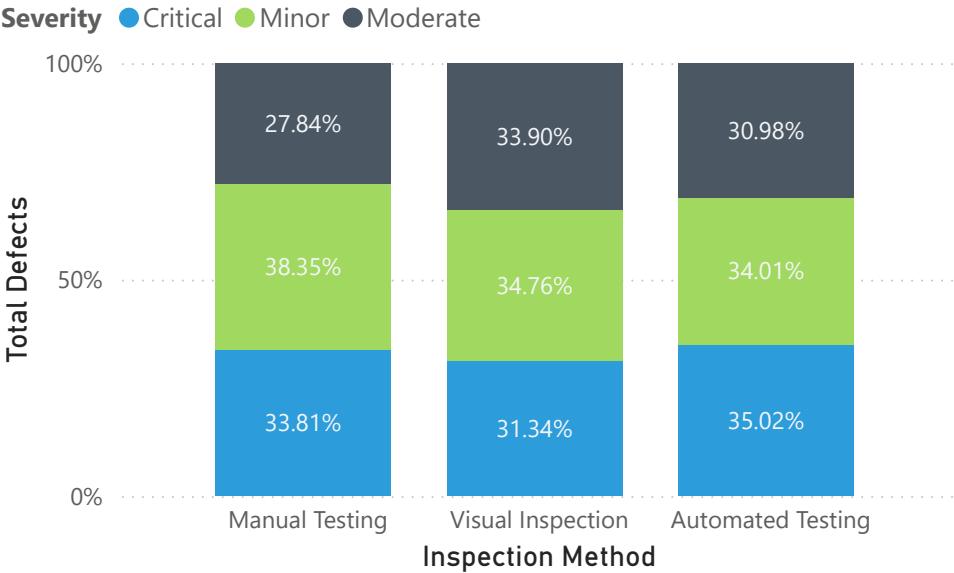
Defect Type Structural Functional Cosmetic



# Inspection Effectiveness

Do different inspection methods surface different operational risks?

## Total Defects by Inspection Method and Severity



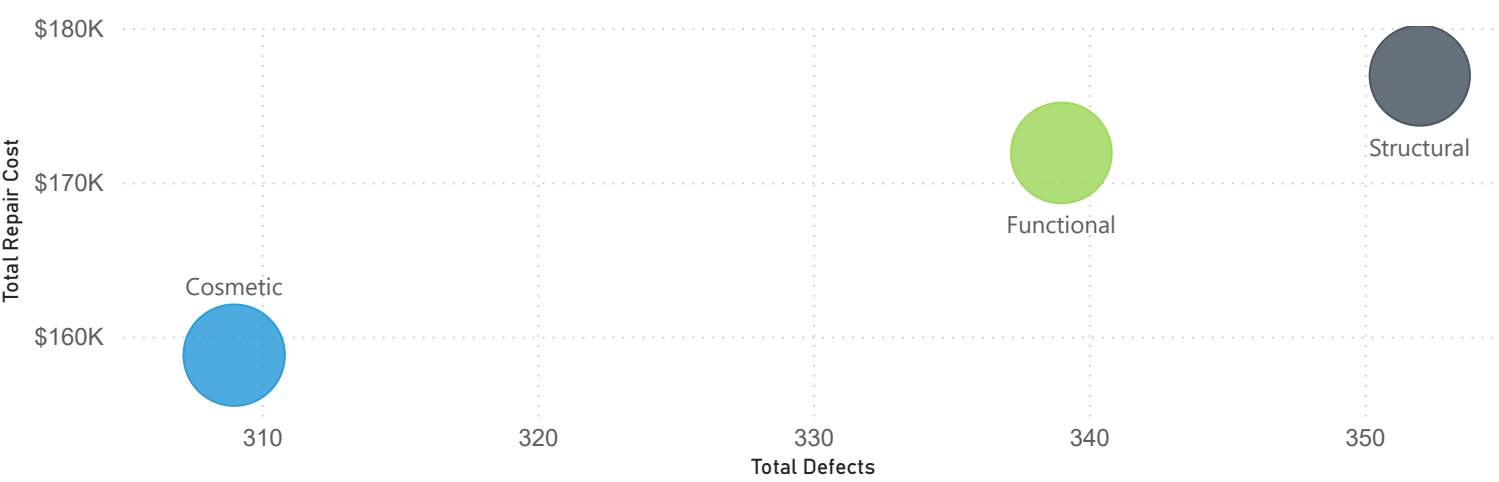
## Avg Repair Cost by Inspection Method and Defect Location

| Inspection Method | Cosmetic | Functional | Structural | Total    |
|-------------------|----------|------------|------------|----------|
| Automated Testing | \$488.93 | \$517.16   | \$498.73   | \$501.80 |
| Manual Testing    | \$530.96 | \$531.98   | \$515.82   | \$526.07 |
| Visual Inspection | \$518.59 | \$471.90   | \$493.02   | \$494.05 |
| Total             | \$513.91 | \$507.10   | \$502.62   | \$507.63 |

# Financial Exposure

Which combinations of defect type, location and severity create the highest financial exposure?

Total Defects, Total Repair Cost and Avg Repair Cost by Defect Type



| Defect Location | Severity | Defect Type | Inspection Method |
|-----------------|----------|-------------|-------------------|
| Component       | Critical | Structural  |                   |

